CIS 4618 – Expert Systems and Decision Support

Decision Table

This activity is designed to help you understand the methodology for building a Decision Table. Decision tables are used to layout in tabular form all possible situations which a business decision may encounter and to specify which action to take in each of these situations.

What is a Decision Table?
A decision table is a tabular form that presents a set of conditions and their corresponding actions.

Parts of a Decision Table
Condition Stubs
Condition stubs describe the conditions or factors that will affect the decision or policy. They are listed in the upper section of the decision table.

Action Stubs
Action stubs describe, in the form of statements, the possible policy actions or decisions. They are listed in the lower section of the decision table.

Rules
Rules describe which actions are to be taken under a specific combination of conditions. They are specified by first inserting different combinations of condition attribute values and then putting X's in the appropriate columns of the action section of the table.

Conditions and Actions	Rules
Conditions	Condition Alternatives
Actions	Action Entries

Figure 1: Parts of Decision Table

Decision Table Methodology

1. Identify Conditions & Values	Find the data attribute each condition tests and all of the attribute's values.
2. Compute Max Number of Rules	Multiply the number of values for each condition data attribute by each other.
3. Identify Possible Action	Determine each independent action to be taken for the decision or policy.
4. Enter All Possible Rules	Fill in the values of the condition data attributes in each numbered rule column.
5. Define Actions for each Rule	For each rule, mark the appropriate actions with an X in the decision table.

Additionally (advised to do), you can simplify the table by eliminating and/or consolidating rules to reduce the number of columns and review completed decision tables with end-users.

Prepared based on Peter Smith tutorial http://www.saintmarys.edu/~psmith/417lab3b.html

Identifying the number of possible rules
The number of rules combination is a product of the number of values for each condition. For example, if there is 1 condition with 3 values and 2 other conditions with 2 values, then there would be 12 combinations (3*2*2=12).

Complete Decision Table Example[1]
Assume that a given Payroll system uses one of several calculations to pay its employees, as expressed in the decision table presented below:

	Conditions/ Courses of Action	Rules					
		1	2	3	4	5	6
Condition Stubs	Employee type	S	H	S	H	S	H
	Hours worked	<40	<40	40	40	>40	>40
Action Stubs	Pay base salary	X		X		X	
	Calculate hourly wage		X		X		X
	Calculate overtime						X
	Produce Absence Report		X				

Legends
S – Salaried Employee
H – Hourly pay Employee

Note: for salaried employees, the action stub chosen will always be the same…therefore hours worked is an indifferent condition

Exercise: Create a decision table for below given medical insurance problem
No charges are reimbursed to the patient until the deductible has been met. Hospital visits are reimbursed at 80%, and Lab visits are reimbursed at 70%. Doctor's office visits are reimbursed at 90% for “Participating Physicians” or 50% for others. The question of whether the Doctor is a Participating Physician is only applicable for Doctor’s office visits; it is not applicable for Hospital visits or Lab work.

The first condition (Is the deductible met?) has two possible outcomes, yes or no. The second condition (the type of visit) has three possible outcomes, Doctor's office visit (D) or Hospital visit (H) or Lab visit (L). There are two possible outcomes to the question of whether in the case of a Doctor's Office visit, the Doctor is a Participating Physician, yes or no. There will be 12 rules (two times three times two is 12).

[1] Based on Modern Systems Analysis and Design book by Hoffer, George and Valacich.

Solution

Conditions	Values	Rules											
		1	2	3	4	5	6	7	8	9	10	11	12
Deductible met?	Y, N	Y	Y	Y	Y	Y	Y	N	N	N	N	N	N
Type of visit	D, H, L	D	D	H	H	L	L	D	D	H	H	L	L
Participating Physician?	Y, N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N
Actions													
Reimburse 50%			X										
Reimburse 70%							X						
Reimburse 80%					X								
Reimburse 90%		X											
No reimbursement								X	X		X		X
Impossible or N/A				X		X				X		X	

Exercise 2: Create a decision table for the below-given problem

A mailing is to be sent out to customers. The content of the mailing is about the current level of discounting and potential levels of discounting. The content is different for different types of customers.

Customer Types A, B, and C get a normal letter except Customer Type C, who get a special letter. Any customer with 2 or more current lines or with a credit rating of 'X' gets a special paragraph added with an offer to subscribe to another level of discounting.

Solution

Conditions/Actions	Values	1	2	3	4	5	6	7	8	9	10	11	12
Conditions													
Customer type	A,B,C	A	A	A	A	B	B	B	B	C	C	C	C
2+ current lines?	Y,N	Y	Y	N	N	Y	Y	N	N	Y	Y	N	N
Credit rating = X?	Y,N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N
Actions													
Send normal letter		X	X	X	X	X	X	X	X				
Send special letter										X	X	X	X
Add special paragraph		X	X	X		X	X	X		X	X	X	